

Clayton W. Ramsey

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Education

Rice University, Houston, TX Aug 2023 - Present
Ph.D., Computer Science (in progress); advised by Dr. Lydia E. Kavraki

Rice University, Houston, TX Aug 2019 - May 2023
B.S., Electrical Engineering; B.A., Computer Science GPA 3.85, cum laude

Experience

Rice University, Houston, TX — *Doctoral Fellow* Aug 2023 - Present

- Proposed coroutine graphs, a massively parallel multithreaded framework for solving task and motion planning in milliseconds. The framework applies insights from operating system scheduling to heuristic search.
- Implemented novel strategies for plan repair, allowing robots to recover from failure in dynamic environments.
- Developed a hardware-accelerated Rust library for sampling-based robot motion planning in arbitrary spaces.
- Proposed CAPT, a SIMD-parallel data structure for fast collision checking with point clouds.

NASA Johnson Space Center, Houston, TX — *Visiting Researcher* May 2025 - Present

- Demonstrated my research in online planning on iMETRO, the robot testing environment at JSC.

Rice University, Houston, TX — *Undergraduate Researcher* May 2021 - Aug 2022

- Implemented a path-plan visualization tool for human verification of motion plans.
- Optimized an algorithm for searching uncertain spaces, yielding a 10x search time reduction.
- Benchmarked skill caching strategies for task planning, yielding up to a 50x speedup in plan generation.

Stellar Solutions, Palo Alto, CA — *Embedded Systems Intern* May 2020 - Aug 2020

- Evaluated a test processor for a satellite payload and developed an SPI driver for external device interfacing.
- Wrote a deserializer converting raw byte data from a ground magnetometer station into usable data, identifying a need for temperature-based sensor calibration.

Stellar Solutions, Palo Alto, CA — *Mechanical Design Intern* Jun 2019 - Aug 2019

- Collaborated with hardware suppliers to advance a design from rough sketch to final model.
- Designed sensor tip for integration with extending boom arm to reduce magnetic interference from onboard electronics.

Selected publications

- **C. W. Ramsey**, Z. Kingston, W. Thomason, L. E. Kavraki. "Collision-Affording Point Trees: SIMD-Amenable Nearest Neighbors for Fast Collision Checking." *Robotics: Science and Systems (RSS)*, 2024.
- **C. W. Ramsey**, L. E. Kavraki. "Coroutine Scheduling in Task and Motion Planning." *IEEE ICRA 2026 Workshop – Robotics Acceleration with Computing Hardware and Systems*, 2026.
- T. Duong, **C. W. Ramsey**, Z. Kingston, W. Thomason, L. E. Kavraki. "Ultrafast Sampling-based Kinodynamic Planning via Differential Flatness." *arXiv preprint arXiv:2603.16059*, 2026.

Awards and honors

- **NSTGRO Fellowship**, NASA, 2024.
- **NDSEG Fellowship**, Department of Defense, 2024.
- **Gold Medalist**, United States Figure Skating, 2018, 2021, and 2024.
- **Eta Kappa Nu Member**, IEEE, 2023.
- **President's Honor Roll**, Rice University, Fall 2022 and Spring 2023.

Skills

- **Programming:** Rust, Python, C, C++, Java, MATLAB, JavaScript, Docker, Pixi
- **Robotics / Research:** task and motion planning, planning under uncertainty, SIMD and GPU hardware acceleration, parallel programming, machine learning